



# Irrigation with simulated oilfield-produced water under semi-arid conditions: growth, seed yield, and oil quality of *Moringa oleifera*

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## Abstract

Water scarcity in semi-arid regions motivates the evaluation of unconventional irrigation sources. This study assessed the growth, seed production, and oil quality of *Moringa oleifera* under irrigation with synthetic saline waters formulated to reproduce the electrical conductivity and major-ion composition of treated oilfield-produced water; no actual produced water was applied. The experiment used a randomized block design in a  $(3 \times 3) + 1$  factorial arrangement, with four replicates. Treatments combined three irrigation levels based on crop evapotranspiration ( $L1=33\%$  ETc,  $L2=66\%$  ETc, and  $L3=100\%$  ETc) and three water electrical conductivities ( $S1=0.5$ ,  $S2=2.5$ , and  $S3=4.5$  dS  $m^{-1}$ ), plus a non-irrigated control. Plant height, stem diameter, dry matter, fruit and seed yield, oil yield, and oil-quality attributes were evaluated. Salinity up to  $4.5$  dS  $m^{-1}$  did not significantly reduce plant height or stem diameter among irrigated treatments, whereas  $100\%$  ETc favored vegetative growth at later evaluations. The highest fruit, seed, and oil yields occurred at  $66\%$  and  $100\%$  ETc, indicating that  $66\%$  ETc can represent a water-saving agronomic compromise when yield is comparable with full irrigation. The  $4.5$  dS  $m^{-1}$  treatment produced the greatest seed and oil yields. These findings demonstrate moringa tolerance to the tested synthetic saline waters under the experimental conditions, but they should not be extrapolated directly to actual produced water because organic contaminants, trace metals, hydrocarbons, and long-term soil salinization or sodicity were not evaluated.

## Introduction

The climate of Brazil's semi-arid region is characterized by seasonal and periodic droughts, which constrain agricultural production and the establishment of plant species

under limited water availability (Lucena et al. 2021; Silva et al. 2024). Climate change may further intensify aridity and increase environmental, economic, and social pressures in the region (Ledru et al. 2020; Soares et al. 2021). Under these conditions, irrigation is essential for many

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crops, yet high-quality freshwater is limited; consequently, localized irrigation and the use of alternative water sources have become increasingly relevant (Cavalcante et al. 2021; Nunes et al. 2022; Silva et al. 2021).

Rio Grande do Norte, Brazil, overlies the Potiguar Basin, an important petroleum-producing region with both onshore and offshore activities (Bezerra et al. 2020). Onshore oil extraction generates substantial volumes of produced water, a complex wastewater whose composition depends on reservoir geology and production and treatment processes (Freitas and Siqueira Filho 2018; Khlaifat et al. 2023). In addition to high salinity, actual produced water may contain residual hydrocarbons, dissolved organic compounds, trace metals, and treatment chemicals. Therefore, any agricultural reuse requires appropriate treatment, water-specific characterization, and environmental monitoring rather than evaluation of salinity alone.

Produced water can contain high concentrations of dissolved salts, and its final composition after treatment varies widely among reservoirs and treatment technologies (Khlaifat et al. 2023; Guo et al. 2023). In water-limited regions, treated produced water could represent a supplementary irrigation source only when its chemical composition complies with applicable environmental and agronomic criteria and when management includes appropriate dilution, drainage, and monitoring of soil and water quality (Saleh et al. 2022). Because real produced water contains constituents beyond the major inorganic ions, experiments using synthetic saline solutions are useful for isolating salinity and ionic-composition effects but cannot reproduce the complete environmental risk profile of the industrial effluent.

Moringa (*Moringa oleifera* Lam.) is a multipurpose oil-seed species capable of growing under limited water availability and other adverse conditions (Ayerza 2019). Its seeds can contain approximately 35–45% oil, with a high proportion of oleic acid, which contributes to oxidative stability and supports its potential use as a biodiesel feedstock (Ayerza 2019; Leone et al. 2016; Omonhinmin et al. 2020; Pereira et al. 2018).

Regarding tolerance to combined water and salt stress, Azeem et al. (2023) reported that *Moringa oleifera* maintained several physiological and biochemical functions under moderate salinity, whereas more severe salinity reduced growth. Under water deficit, moringa can also respond through stomatal regulation, osmotic adjustment, and enhanced antioxidant metabolism (Brunetti et al. 2018; Azeem et al. 2023). These responses make the species a relevant model for evaluating deficit irrigation with moderately saline water in semi-arid environments.

Accordingly, we hypothesized that *Moringa oleifera* could maintain growth and oil production when irrigated with synthetic saline water formulated to reproduce the

electrical conductivity and major-ion composition of treated oilfield-produced water. Importantly, the experiment did not use actual produced water and therefore did not evaluate residual hydrocarbons, trace metals, or other organic and inorganic contaminants that may remain after industrial treatment. The objective was to determine the effects of irrigation level and simulated produced-water salinity on vegetative growth, seed and fruit yield, oil yield, and crude-oil quality under semi-arid field conditions.

## Materials and methods

The experiment was conducted from July 2022 to December 2023 in an experimental area on the East Campus of the Federal Rural University of the Semi-Arid (UFERSA), Center for Agricultural Sciences, Mossoró, Rio Grande do Norte, Brazil. The site is located at 5°12'48" S, 37°18'44" W, at an altitude of 37 m above sea level (Fig. 1).

The climate is classified as BSwH' according to Köppen, characterized by high temperatures, scarce and irregular rainfall, and a marked dry season (Alvares et al. 2013). Mean air temperature (T<sub>med</sub>), reference evapotranspiration (ET<sub>o</sub>), global radiation (R<sub>g</sub>), and relative humidity (RH) were recorded daily, as shown in Fig. 2.

Before the field experiment, disturbed soil samples were collected with a Dutch-type auger (EQUILAB, Uberaba, MG, Brazil) at depths of 0.0–0.2 and 0.2–0.4 m. Samples from each depth were homogenized to obtain composite samples representative of the area and were analyzed for physical and chemical properties at the Semi-Arid Soil, Water, and Plant Analysis Laboratory (LASAPSA). The soil was classified as an Argissolo, corresponding to an Ultisol in Soil Taxonomy, and was initially non-sodic and non-saline to a depth of 0.4 m (Richards 1969; Santos et al. 2018; Soil Survey Staff 2022) (Table 1).

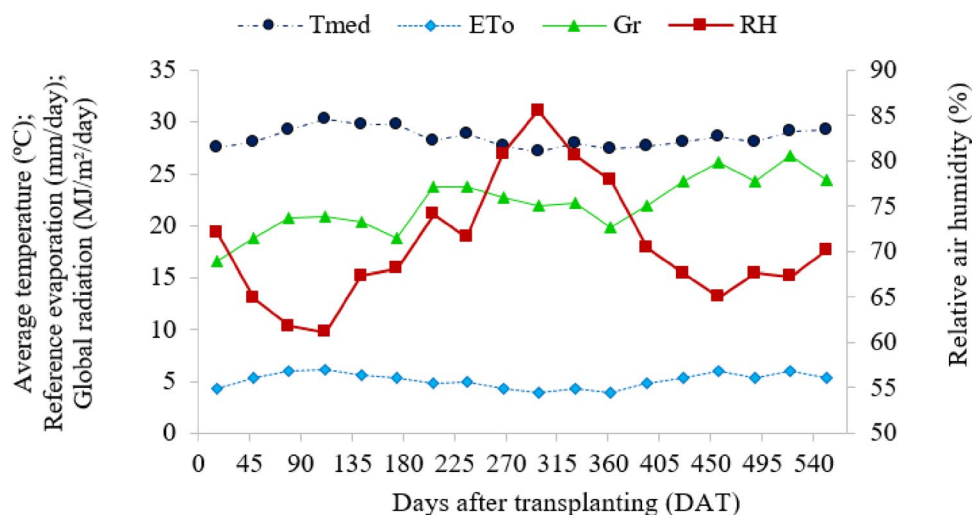
The experiment was arranged in a randomized complete block design in a (3×3)+1 factorial scheme, with four replicates. Each experimental plot consisted of a row of six plants, of which the three central plants were used for evaluation (Fig. 3). Treatments combined three irrigation levels based on crop evapotranspiration (ET<sub>c</sub>: L1=33%, L2=66%, and L3=100% ET<sub>c</sub>) and three levels of irrigation-water electrical conductivity (S1=0.5, S2=2.5, and S3=4.5 dS m<sup>-1</sup>), plus a control treatment (T0) in which irrigation was discontinued 60 days after transplanting (DAT).

Seeds used for seedling production were collected from parent plants on the UFERSA East Campus. Seedlings were produced in 200-cell polyethylene trays with one seed per cell. The substrate consisted of a 1:1 mixture of commercial Coco<sup>®</sup> and Carolina<sup>®</sup> substrates; the latter contained peat, vermiculite, carbonized rice husk, organic residues, and an



**Fig. 1** Aerial view and experimental layout of the *Moringa oleifera* cultivation, showing the overall plot arrangement. Photograph courtesy of Francielle Gurgel de Castro Alves. Copyright 2023

**Fig. 2** Meteorological conditions during the experimental period



NPK 10-10-10 formulation. Seedlings were irrigated manually with local supply water ( $EC_w = 0.5 \text{ dS m}^{-1}$ ). Emergence began approximately 5 days after sowing (DAS), and seedlings were transplanted at 30 DAS.

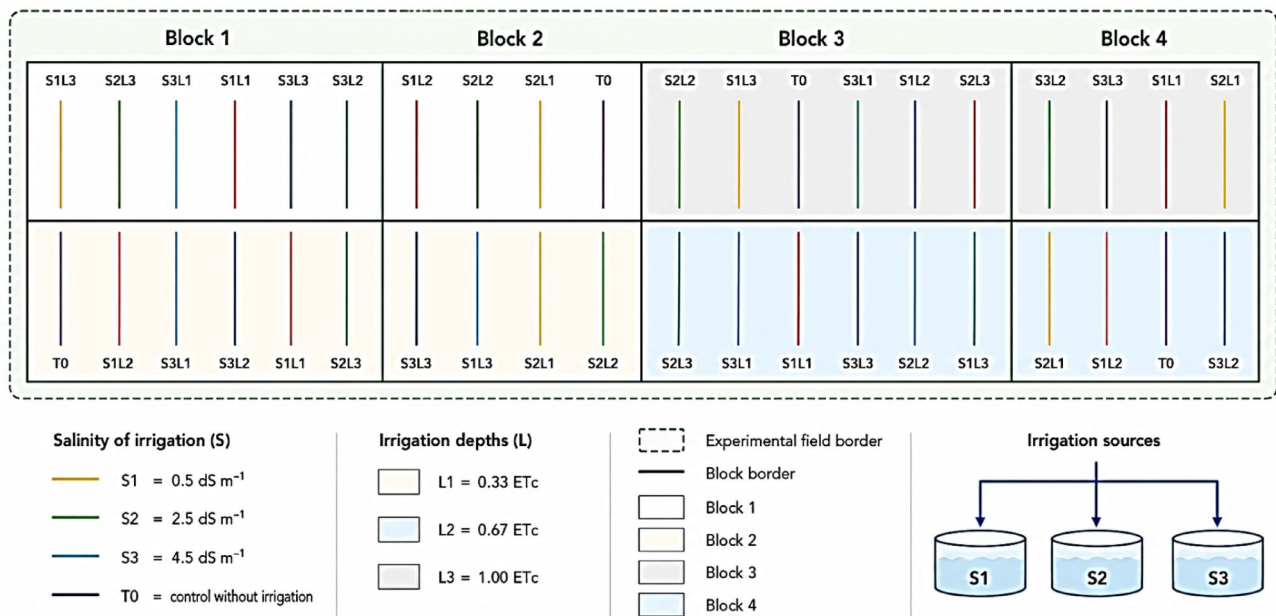
Before transplanting, the upper 0.0–0.2 m soil layer was harrowed. Planting holes ( $0.50 \times 0.50 \times 0.40 \text{ m}$ ) were opened at  $3.0 \times 1.5 \text{ m}$  spacing and fertilized with 50 g of

monoammonium phosphate (47%  $P_2O_5$  and 5% N) per hole. One seedling was transplanted per hole. Pruning was performed at 90, 150, and 270 DAT. At 270 DAT, alternate plants were drastically pruned so that three central plants remained for evaluation, increasing light penetration and lateral canopy development and resulting in a final 3.0 m spacing between evaluated plants.

**Table 1** Initial chemical and physical characterization of the soil before establishment of the experiment

| Chemical properties                 | Unit                              | 0.0–0.2 m | 0.2–0.4 m | Physical properties | Unit               | 0.0–0.2 m | 0.2–0.4 m |
|-------------------------------------|-----------------------------------|-----------|-----------|---------------------|--------------------|-----------|-----------|
| pH (H <sub>2</sub> O)               | –                                 | 7.62      | 7.66      | Coarse sand         | g kg <sup>-1</sup> | 5.4       | 4.7       |
| EC                                  | dS m <sup>-1</sup>                | 0.19      | 0.18      | Fine sand           | g kg <sup>-1</sup> | 2.4       | 2.4       |
| K <sup>+</sup>                      | cmol <sub>c</sub> L <sup>-1</sup> | 0.37      | 0.38      | Total sand          | g kg <sup>-1</sup> | 7.8       | 7.1       |
| Ca <sup>2+</sup> + Mg <sup>2+</sup> | cmol <sub>c</sub> L <sup>-1</sup> | 1.91      | 1.97      | Silt                | g kg <sup>-1</sup> | 1.3       | 1.3       |
| Na <sup>+</sup>                     | cmol <sub>c</sub> L <sup>-1</sup> | 0.35      | 0.37      | Clay                | g kg <sup>-1</sup> | 0.9       | 1.7       |
| Al <sup>3+</sup>                    | cmol <sub>c</sub> L <sup>-1</sup> | 0         | 0         | –                   | –                  | –         | –         |
| Al <sup>3+</sup> + H <sup>+</sup>   | cmol <sub>c</sub> L <sup>-1</sup> | 0         | 0         | –                   | –                  | –         | –         |
| SB                                  | cmol <sub>c</sub> L <sup>-1</sup> | 2.63      | 2.72      | –                   | –                  | –         | –         |
| CEC                                 | cmol <sub>c</sub> L <sup>-1</sup> | 2.63      | 2.72      | –                   | –                  | –         | –         |
| V                                   | %                                 | 100       | 100       | –                   | –                  | –         | –         |

\*pH (Hydrogen potential): Measured using a Tecnal pH meter (Model Tec-2), Piracicaba-SP, Brazil. EC (Electrical conductivity): Determined with a digital bench conductivity meter (Model CD-820, Instrutherm), São Paulo-SP, Brazil. K<sup>+</sup> (Potassium): Quantified by digital flame photometry (Werberlab brand), SC/PR, Brazil. Ca<sup>2+</sup> (Calcium): Determined according to the methodology described by Teixeira et al. (2017). Mg<sup>2+</sup> (Magnesium): Analyzed concurrently with calcium. Na<sup>+</sup> (Sodium): Quantified by digital flame photometry (Werberlab brand), SC/PR, Brazil. Al<sup>3+</sup> (Aluminum): Extracted and titrated. Al<sup>3+</sup> + H<sup>+</sup> (Soil acidity): Represents potential soil acidity



**Fig. 3** Field layout of the randomized complete block design. Each of the four blocks contained all combinations of three simulated-water salinity levels (S1=0.5 dS m<sup>-1</sup>; S2=2.5 dS m<sup>-1</sup>; S3=4.5 dS m<sup>-1</sup>) and

three irrigation levels (L1=0.33 ETc; L2=0.66 ETc; L3=1.00 ETc), plus the non-irrigated control (T0)

Irrigation volume was determined from the water balance in the crop root zone (to 0.55 m). Changes in water storage and drainage below the root system were considered negligible for the irrigation calculation because the underlying soil layer had greater clay content and lower hydraulic conductivity, with moisture remaining below field capacity (Connolly 1998).

Soil-water retention curves were determined using a tension-table method. Volumetric water contents at field capacity in the 0–0.2, 0.2–0.4, and 0.4–0.6 m layers were 0.225, 0.234, and 0.236 m<sup>3</sup> m<sup>-3</sup>, corresponding to tensions of 2.2, 1.9, and 2.3 kPa, respectively. Water contents at the

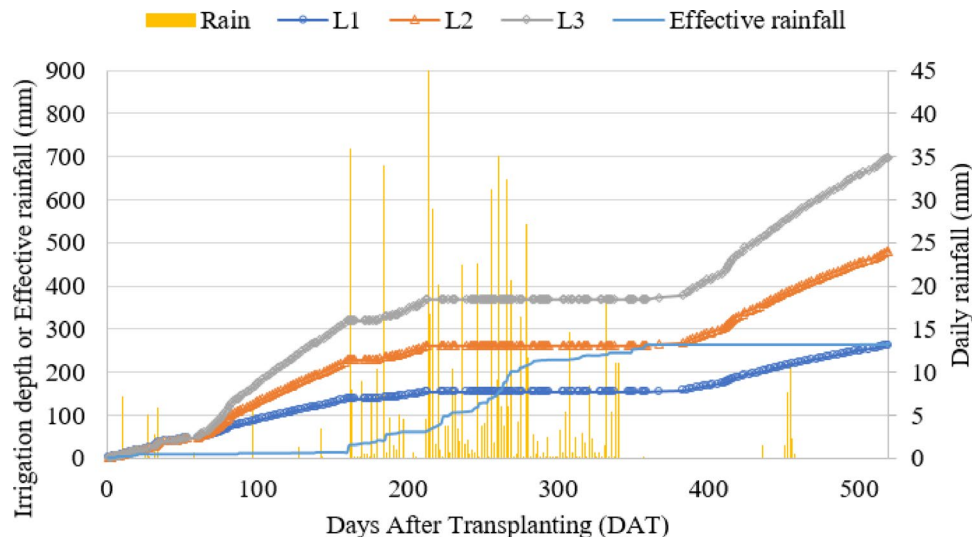
permanent wilting point were 0.031, 0.084, and 0.093 m<sup>3</sup> m<sup>-3</sup>. Critical soil-water tensions were 12.5, 10.6, and 18.2 kPa, and the readily available water fraction was set at 0.5, following Santos et al. (2017) for moringa.

Crop water status was monitored using three tensiometers installed at depths of 0.15, 0.35, and 0.55 m in the plot receiving 100% ETc, which served as the reference for irrigation scheduling. The mean 30-day water balance was calculated from the irrigation depth applied to this treatment and a localized-irrigation efficiency factor of 0.90, following Reichard and Timm (2022) (Eq. 1):

**Table 2** Chemical characterization of the synthetic saline waters formulated to simulate the major-ion composition of treated oilfield-produced water used for moringa irrigation

| Water | EC (dS m <sup>-1</sup> ) | Na <sup>+</sup> (g L <sup>-1</sup> ) | Ca <sup>2+</sup> (g L <sup>-1</sup> ) | Mg <sup>2+</sup> (g L <sup>-1</sup> ) | Cl <sup>-</sup> (g L <sup>-1</sup> ) | SO <sub>4</sub> <sup>2-</sup> (g L <sup>-1</sup> ) | HCO <sub>3</sub> <sup>-</sup> (g L <sup>-1</sup> ) |
|-------|--------------------------|--------------------------------------|---------------------------------------|---------------------------------------|--------------------------------------|----------------------------------------------------|----------------------------------------------------|
| S1    | 0.5                      | 2.5                                  | 1.5                                   | 1.0                                   | 1.3                                  | 0.2                                                | 3.5                                                |
| S2    | 2.5                      | 12.5                                 | 6.5                                   | 3.5                                   | 15.0                                 | 1.0                                                | 6.5                                                |
| S3    | 4.5                      | 25.0                                 | 13.0                                  | 7.0                                   | 37.0                                 | 2.0                                                | 6.0                                                |

\*ECw Water electrical conductivity, Na<sup>+</sup> Sodium, Ca<sup>2+</sup> Calcium, Mg<sup>2+</sup> Magnesium, Cl<sup>-</sup> Chloride, SO<sub>4</sub><sup>2-</sup> Sulfate anion, HCO<sub>3</sub><sup>-</sup> Bicarbonate

**Fig. 4** Cumulative irrigation depth applied to treatments L1, L2, and L3, daily rainfall, and cumulative estimated effective rainfall during the experimental period

$$ETc = \text{Irrigation (day)} \times 0,9 \quad (1)$$

From transplanting to 60 DAT, all treatments received the same localized irrigation through drip tape with emitters spaced 0.5 m apart and a nominal flow rate of 90 L h<sup>-1</sup> per plant. Water was supplied by the Rio Grande do Norte Water and Sewage Company (CAERN; ECw=0.5 dS m<sup>-1</sup>; SAR=2.23), with three 10-min irrigation events per week. From 90 DAT onward, treatments were differentiated using microtubes with one emitter per plant and treatment-specific flow control obtained by adjusting emitter diameter and length (L1=30 L h<sup>-1</sup>, L2=60 L h<sup>-1</sup>, and L3=90 L h<sup>-1</sup>).

The S1 water (0.5 dS m<sup>-1</sup>) was obtained from the urban supply system (CAERN). The S2 and S3 waters (2.5 and 4.5 dS m<sup>-1</sup>, respectively) were prepared synthetically by adding NaCl, CaCl<sub>2</sub>·2 H<sub>2</sub>O, and a concentrated mother solution containing Mg<sup>2+</sup>, Na<sup>+</sup>, and K<sup>+</sup> to S1 water. The formulation was adjusted to obtain a final molar charge ratio of 7:2:1 (Na: Ca: Mg). Electrical conductivity was verified with a CD-820 bench conductivity meter (Instrutherm, São Paulo, Brazil).

The salt formulation was designed to simulate the salinity and major-ion proportions reported for treated produced water from oil reservoirs in Rio Grande do Norte (Souza 2023). Thus, S2 and S3 were synthetic saline waters and not actual oilfield-produced water. They reproduced the targeted electrical conductivity and major inorganic-ion composition only; residual hydrocarbons, dissolved organic

compounds, trace metals, treatment chemicals, and other constituents that may occur in real produced water were not included. This distinction limits direct extrapolation of the results to field reuse of actual produced water. Before irrigation, each water was characterized for electrical conductivity and major-ion composition (Table 2).

Irrigation-water electrical conductivity was measured with a CD-820 bench conductivity meter (Instrutherm, São Paulo, Brazil), and sodium was determined by flame photometry (Werberlab, Brazil). Calcium, magnesium, and chloride were analyzed according to Teixeira et al. (2017). Treatments were applied during two dry-season periods, from 60 to 210 DAT and from 390 to 525 DAT, totaling 275.2 mm, 504.6 mm, and 733.9 mm of irrigation water in L1, L2, and L3, respectively. During the rainy season, 708 mm of effective rainfall accumulated and irrigation was not required (Fig. 4).

Biometric measurements were made at 90, 150, 270, 450, and 525 DAT. Plant height was measured from the soil surface to the apical bud. Stem diameter was measured with a digital caliper (0.1-mm resolution; MTX 6 POL, 150 mm); when stem size exceeded the caliper range, stem circumference was measured with a tape and converted to diameter using Eq. 2:

$$L = \pi \times SD \quad (2)$$

Where:

$L$ =Length;  
 $\pi$ =3.1416; and  
 $SD$ =Stem diameter.

Shoot dry matter (DM; stem+leaves) was determined after cutting plants at 1.20 m with a portable pruning saw and recording fresh mass with a 50-kg-capacity digital hanging scale. The material was then processed in a forage chopper, subsampled into kraft-paper bags, and dried in a forced-air oven (TE-394/2-MP, Tecnal®, Brazil) at 65 °C for 72 h. Dry-matter proportion was calculated from the ratio between oven-dry and fresh mass (AOAC 2009).

Fruits were harvested manually between 475 and 525 DAT using pruning shears. The total number of capsules was recorded for each experimental unit. Capsules were then threshed, seeds were counted, and fruit and seed production variables were calculated according to Eqs. 3 and 4.

$$\text{Capsule yield (\%)} = \frac{\text{No. of capsules per treatment}}{\text{Cumulative capsule yield per cycle}} \quad (3)$$

$$\text{Fruit yield (\%)} = \frac{\text{No. of fruit per capsule}}{\text{Cumulative fruit yield per cycle}} \quad (4)$$

Oil extraction and quality analyses were performed at the UFERSA Petroleum and Biofuels Research Laboratory. After seed counting, the seed coats were removed and kernels were ground in a Wiley-type knife mill. A 20-g ground sample was extracted with 160 mL hexane for 4 h in a Soxhlet apparatus (AOAC 2009). Hexane was removed with a rotary evaporator (Fitason 558, Rio de Janeiro, Brazil), and the extracted oil was filtered and subsequently

heated in a water bath at 105 °C (AOAC 2009). Oil yield was calculated using Eq. 5:

$$\text{Oil yield (\%)} = \frac{\text{Extracted Oil Weight (g)}}{\text{initial sample weight (g)}} \quad (5)$$

Crude-oil quality was evaluated according to AOCS procedures (AOCS 1985). The acidity index was determined by titration with 0.1 N sodium hydroxide using phenolphthalein as indicator, following Instituto Adolfo Lutz (2008), and expressed as mg KOH g<sup>-1</sup> oil. Turbidity was measured with a portable digital turbidimeter (Akron KR2000, São Leopoldo, Rio Grande do Sul, Brazil). Kinematic viscosity and density were measured simultaneously from a 5.0-mL crude-oil sample using an automatic kinematic viscometer (Anton Paar SVM 1101, Caratinga, Minas Gerais, Brazil).

Homogeneity of residual variances and normality were evaluated using Bartlett's and Shapiro-Wilk tests, respectively. Analysis of variance was then performed using the F test ( $p \leq 0.05$ ). Treatment means were compared with the non-irrigated control using Dunnett's test at 5% probability, and the isolated effects of irrigation level or salinity were compared using Tukey's test at 5% probability. Analyses were performed in R version 4.3.1 in the RStudio environment.

## Results and discussion

There was no significant salinity  $\times$  irrigation-level interaction ( $p > 0.05$ ) for plant height during the evaluated periods (Table 3). Salinity alone did not significantly affect height, whereas irrigation level affected height at 450 and 525 DAT. Comparisons between the irrigated treatments and the non-irrigated control showed a clear benefit of water supply during the dry-season evaluations. Therefore, the data support tolerance to the tested salinity range under irrigation, rather than a general conclusion that produced-water irrigation is intrinsically beneficial.

Water stress limits plant growth and biomass through profound physiological and biochemical changes that minimize damage caused by water shortage in tissues. In addition, it can increase secondary metabolites in leaves (such as phenolic compounds) and reduce these compounds in roots, prioritizing their defense and survival (Mundim and Pringle 2018; Vijayaragavan et al. 2025; Wu et al. 2022).

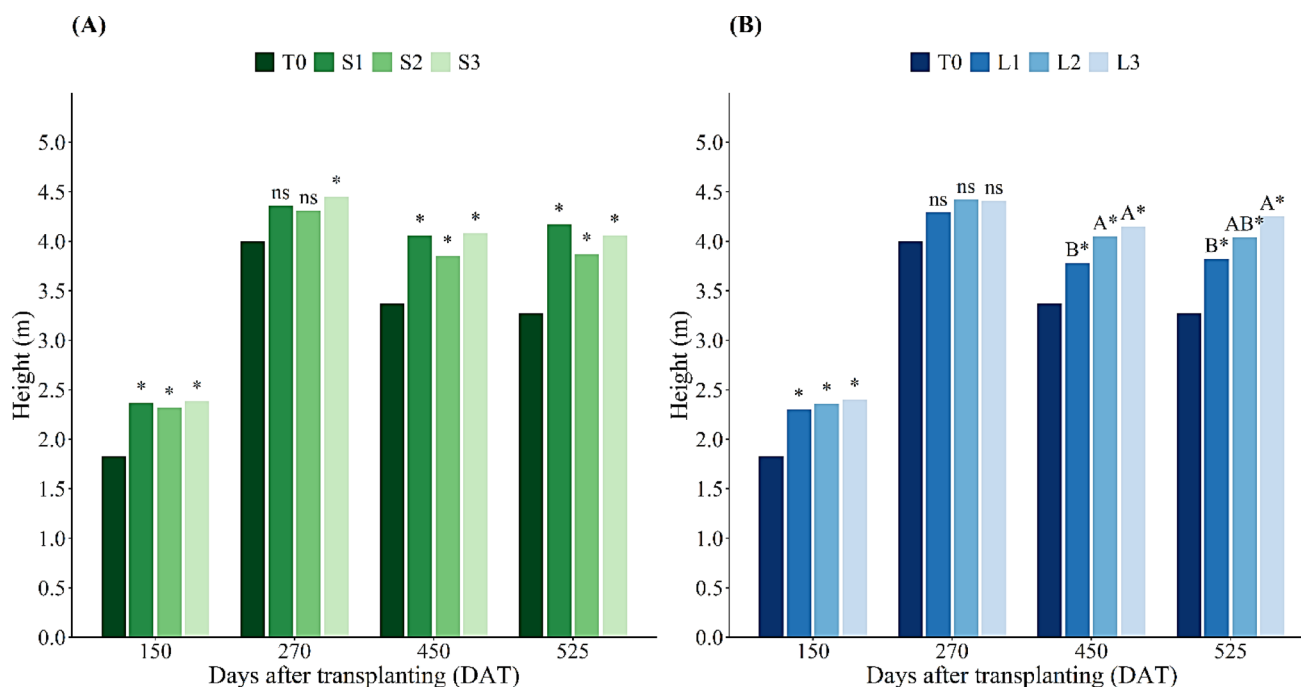
Salinity did not significantly affect plant height among the irrigated treatments (Fig. 5A). Nevertheless, mean height under irrigation was greater than in the non-irrigated control during the treatment-application periods, with increases of 29.04%, 18.6%, and 23.5% at 150, 450, and 525 DAT, respectively. At 270 DAT, differences were attenuated

**Table 3** Analysis of variance for moringa plant height at different evaluation periods under irrigation levels (Irr) and salinity levels (Sal) of simulated oilfield-produced water

| SV                              | DF    | Height (m)          |                     |                     |                     |
|---------------------------------|-------|---------------------|---------------------|---------------------|---------------------|
|                                 |       | 150 DAT             | 270 DAT             | 450 DAT             | 525 DAT             |
|                                 |       | F-test significance |                     |                     |                     |
| Fat × T0                        | 1     | 0.000**             | 0.034*              | 0.000**             | 0.000**             |
| Sal                             | 2     | 0.686 <sup>ns</sup> | 0.517 <sup>ns</sup> | 0.054 <sup>ns</sup> | 0.060 <sup>ns</sup> |
| Bla                             | 2     | 0.355 <sup>ns</sup> | 0.538 <sup>ns</sup> | 0.003**             | 0.005**             |
| Sal × Blz                       | 4     | 0.447 <sup>ns</sup> | 0.421 <sup>ns</sup> | 0.072 <sup>ns</sup> | 0.324 <sup>ns</sup> |
| dms (Sal or Bla) Tukey          | 17.8  | 31.7                | 25.1                | 30.2                |                     |
| Dms (Trat) Tukey                | 30.9  | 54.99               | 43.4                | 52.38               |                     |
| dms (Dunnett) T0 vs. Sal or Bla | 23.7  | 42.2                | 33.3                | 40.2                |                     |
| dms (Dunnett) T0 vs Trat        | 35.3  | 62.92               | 49.66               | 59.94               |                     |
| CV (%)                          | 24    | 7.6                 | 6.3                 | 7.5                 |                     |
| Means                           | 122.6 | 230.2               | 393.1               | 396                 |                     |

SV Source of variation, DF Degrees of freedom, DAT Days after transplanting, Fat Means of irrigated treatments, LSD Least significant difference, CV Coefficient of variation, T0 Non-irrigated control treatment, ns Not significant

\*Significant at the 5% probability level; \*\*Significant at the 1% probability level



**Fig. 5** Moringa plant height at different evaluation periods as affected by salinity of simulated oilfield-produced water (A) and irrigation level (B). \*Significant difference from the non-irrigated control (T0) by Dunnett's test at 5% probability. Identical lowercase letters indicate

no significant difference among salinity levels, and identical uppercase letters indicate no significant difference among irrigation levels by Tukey's test ( $p \leq 0.05$ )

because this evaluation occurred during the rainy season, when treatments had not been applied since 210 DAT.

The absence of a significant reduction in height up to 4.5 dS m<sup>-1</sup> likely reflects a combination of moringa's capacity to tolerate moderate salinity and the environmental conditions of the experiment rather than an absence of salt stress. *Moringa oleifera* can respond to salinity through osmotic adjustment and antioxidant defenses that help maintain cell water status and limit oxidative damage (Azeem et al. 2023; Bayomy et al. 2025). The presence of Ca<sup>2+</sup> and Mg<sup>2+</sup> in the synthetic waters may also have moderated ionic imbalance relative to NaCl-only salinity. In addition, the soil initially had low electrical conductivity (Table 1), and 708 mm of effective rainfall during the rainy season may have redistributed or leached part of the accumulated salts. Because tissue ion concentrations, osmotic adjustment, antioxidant activity, and post-experiment soil salinity were not measured, these mechanisms should be regarded as plausible explanations rather than directly demonstrated processes.

At 450 and 525 DAT, 100% ETc (L3) increased plant height by 9.68% and 11.6%, respectively, relative to 66% ETc (L2), and by 7.30% and 5.76% relative to 33% ETc (L1) (Fig. 5B). Thus, full irrigation favored vegetative growth at later stages. This response is physiologically consistent with greater water availability maintaining cell expansion and carbon assimilation, whereas deficit irrigation requires

greater stomatal regulation and resource allocation to stress acclimation.

At 150 DAT, plant height was 25.9%, 29.42%, and 31.73% greater in L1, L2, and L3, respectively, than in the non-irrigated control; at 525 DAT, the corresponding increases were 16.77%, 23.50%, and 30.32%. The 270-DAT evaluation occurred during the rainy season, when rainfall reduced differences in water availability among treatments. This temporary convergence is compatible with short-term recovery after water deficit, whereas persistent drought can reduce organ expansion through stomatal limitation, altered biomass partitioning, and osmotic adjustment (Seleiman et al. 2021).

Table 4 summarizes the analysis of variance for stem diameter. No significant salinity  $\times$  irrigation-level interaction or isolated salinity effect was detected ( $p > 0.05$ ). Irrigation level affected stem diameter at 180, 480, and 555 DAT, and irrigated treatments differed from the control at those same evaluations. At 120 DAT, no treatment effect was detected, and mean stem diameter was 24.4 mm.

Stem diameter did not differ significantly among salinity levels at the evaluated times (Fig. 6A). Relative to the non-irrigated control, however, irrigated plants had larger stem diameters at 180, 480, and 555 DAT. Therefore, within the tested range, increasing the electrical conductivity of the synthetic water to 4.5 dS m<sup>-1</sup> did not produce a detectable additional reduction in stem diameter. This result should

**Table 4** Analysis of variance for moringa stem diameter at different evaluation periods under irrigation levels (Irr) and salinity levels (Sal) of simulated oilfield-produced water

| FV                              | DF | Diameter (mm)       |                     |                     |                     |
|---------------------------------|----|---------------------|---------------------|---------------------|---------------------|
|                                 |    | 120<br>DAT          | 180<br>DAT          | 480<br>DAT          | 555<br>DAT          |
|                                 |    | F-test significance |                     |                     |                     |
| Fat × T0                        | 1  | 0.060 <sup>ns</sup> | 0.000**             | 0.000**             | 0.003**             |
| Sal                             | 2  | 0.458 <sup>ns</sup> | 0.487 <sup>ns</sup> | 0.335 <sup>ns</sup> | 0.114 <sup>ns</sup> |
| Bla                             | 2  | 0.799 <sup>ns</sup> | 0.005**             | 0.029*              | 0.036*              |
| Sal × Bla                       | 4  | 0.995 <sup>ns</sup> | 0.919 <sup>ns</sup> | 0.599 <sup>ns</sup> | 0.873 <sup>ns</sup> |
| Dms (Sal or Bla) Tukey          |    | 5.71                | 6.47                | 9.97                | 12.54               |
| Dms (Trat) Tukey                |    | 9.89                | 11.20               | 17.27               | 21.73               |
| Dms (Dunnett) T0 vs. Sal or Bla |    | 7.60                | 8.61                | 13.27               | 16.69               |
| dms (Dunnett) T0 vs Trat        |    | 11.32               | 12.82               | 19.76               | 24.86               |
| CV (%)                          |    | 23.1                | 10.0                | 7.3                 | 8.7                 |
| Means                           |    | 24.4                | 63.8                | 134.7               | 142.1               |

SV Source of variation, DF Degrees of freedom, DAT Days after transplanting, Fat Means of irrigated treatments; LSD Least significant difference, CV Coefficient of variation, T0 Non-irrigated control treatment, ns Not significant

\*Significant at the 5% probability level; \*\*Significant at the 1% probability level

not be interpreted as evidence that actual produced water is harmless, because the synthetic solutions did not contain the non-saline contaminants that may occur in field effluents.

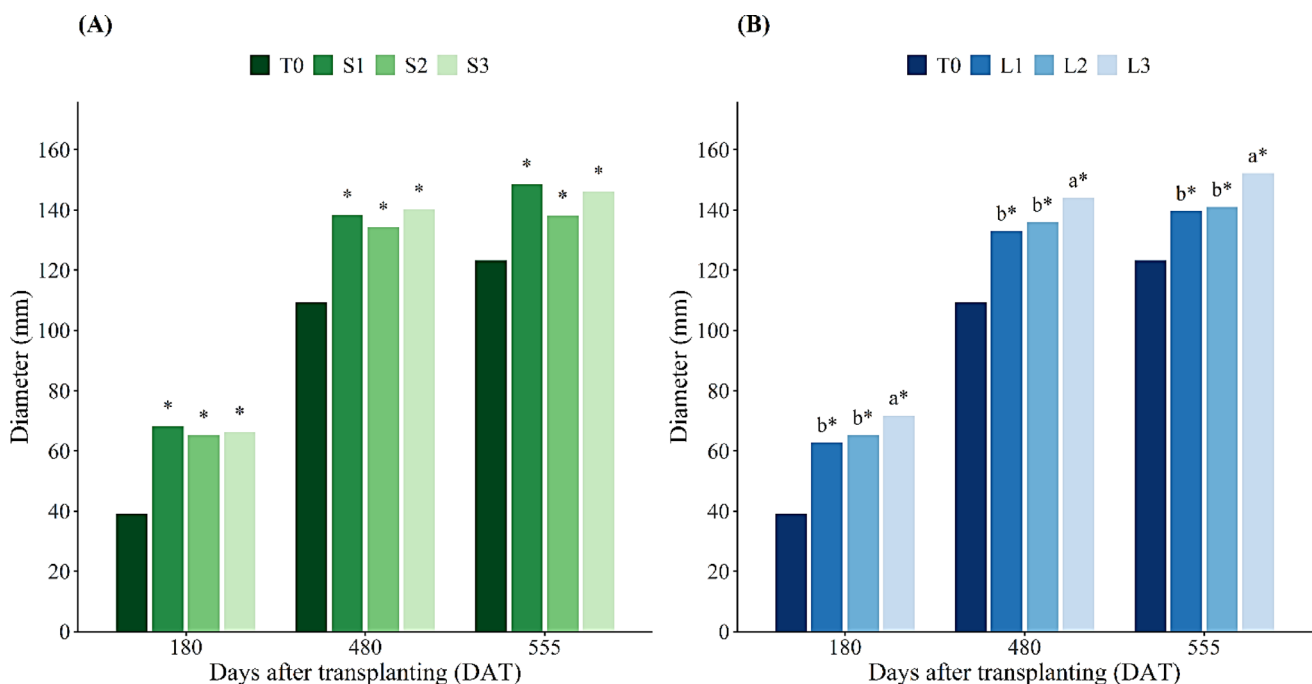
As shown in Fig. 6B, reducing irrigation from 100% ETc (L3) to 66% ETc (L2) decreased stem diameter by 10.11%,

5.99%, and 8.01% at 180, 480, and 555 DAT, respectively. Under the more restrictive 33% ETc treatment (L1), reductions relative to L3 were 14.43%, 8.32%, and approximately 9–10% over the same evaluations. Despite these reductions, all irrigated treatments maintained larger stems than the non-irrigated control during the dry-season assessments, confirming that water availability exerted a stronger effect on this vegetative trait than the tested salinity range.

The coordinated reduction in stem diameter and plant height under severe water restriction is consistent with the limitation of cell expansion and growth under low plant water status. Hajaji et al. (2024) similarly observed marked reductions in height and diameter of moringa subjected to severe water restriction, reflecting a conservative growth strategy under drought.

Shoot dry matter differed significantly between irrigated treatments and the non-irrigated control in all evaluated periods (Table 5). Even 33% ETc maintained greater biomass accumulation than the absence of irrigation during the dry season, demonstrating the agronomic value of supplemental water for moringa under semi-arid conditions. This result supports the use of deficit irrigation as a water-saving strategy but does not, by itself, establish the safety of using actual oilfield-produced water.

At 525 DAT, dry matter under the 0.5 dS m<sup>-1</sup> treatment exceeded that under 2.5 dS m<sup>-1</sup> and was similar to that under 4.5 dS m<sup>-1</sup>. The lack of a monotonic salinity response may



**Fig. 6** Moringa stem diameter at different evaluation periods as affected by salinity of simulated oilfield-produced water (A) and irrigation level (B). \*Significant difference from the non-irrigated control (T0) by Dunnett's test at 5% probability. Identical lowercase letters

indicate no significant difference among salinity levels, and identical uppercase letters indicate no significant difference among irrigation levels by Tukey's test ( $p \leq 0.05$ )

**Table 5** Analysis of variance for moringa shoot dry matter at different evaluation periods under irrigation levels (Irr) and salinity levels (Sal) of simulated oilfield-produced water

| FV                               | DF | Dry matter (DM)       |                     |                     |                     |
|----------------------------------|----|-----------------------|---------------------|---------------------|---------------------|
|                                  |    | 90 DAT                | 150 DAT             | 270 DAT             | 525 DAT             |
|                                  |    | F - Test significance |                     |                     |                     |
| Fat × T0                         | 1  | 0.001**               | 0.000**             | 0.001**             | 0.010**             |
| Sal                              | 2  | 0.561 <sup>ns</sup>   | 0.363 <sup>ns</sup> | 0.433 <sup>ns</sup> | 0.019*              |
| Bla                              | 2  | 0.0001**              | 0.061 <sup>ns</sup> | 0.204 <sup>ns</sup> | 0.067 <sup>ns</sup> |
| Sal × Bla                        | 4  | 0.557 <sup>ns</sup>   | 0.522 <sup>ns</sup> | 0.340 <sup>ns</sup> | 0.652 <sup>ns</sup> |
| dms(Sal or Bla) Tukey            |    | 234.8                 | 402.4               | 1930.9              | 1012                |
| dms(Trat) Tukey                  |    | 406.67                | 697.03              | 3344.49             | 1752.9              |
| dms(Dunnett)T0 vs. Sal<br>or Bla |    | 312.4                 | 535.4               | 2569.1              | 1346.5              |
| dms (Dunnett)To vs.<br>Trat      |    | 465.34                | 797.58              | 3826.98             | 2005.78             |
| CV (%)                           |    | 46.4                  | 30.4                | 27.4                | 48.4                |
| Means                            |    | 499.4                 | 1305.5              | 6966.8              | 2061.5              |

FV Source of variation, DF Degrees of freedom, DAT Days after transplanting; Fat Means of irrigated treatments; LSD Least significant difference; CV Coefficient of variation; T0 Non-irrigated control treatment; ns Not significant

\*Significant at the 5% probability level; \*\*Significant at the 1% probability level

reflect concurrent reproductive allocation because this evaluation coincided with fruit development and capsule harvest. Biomass partitioning toward reproductive organs can reduce contemporaneous vegetative dry matter. However,

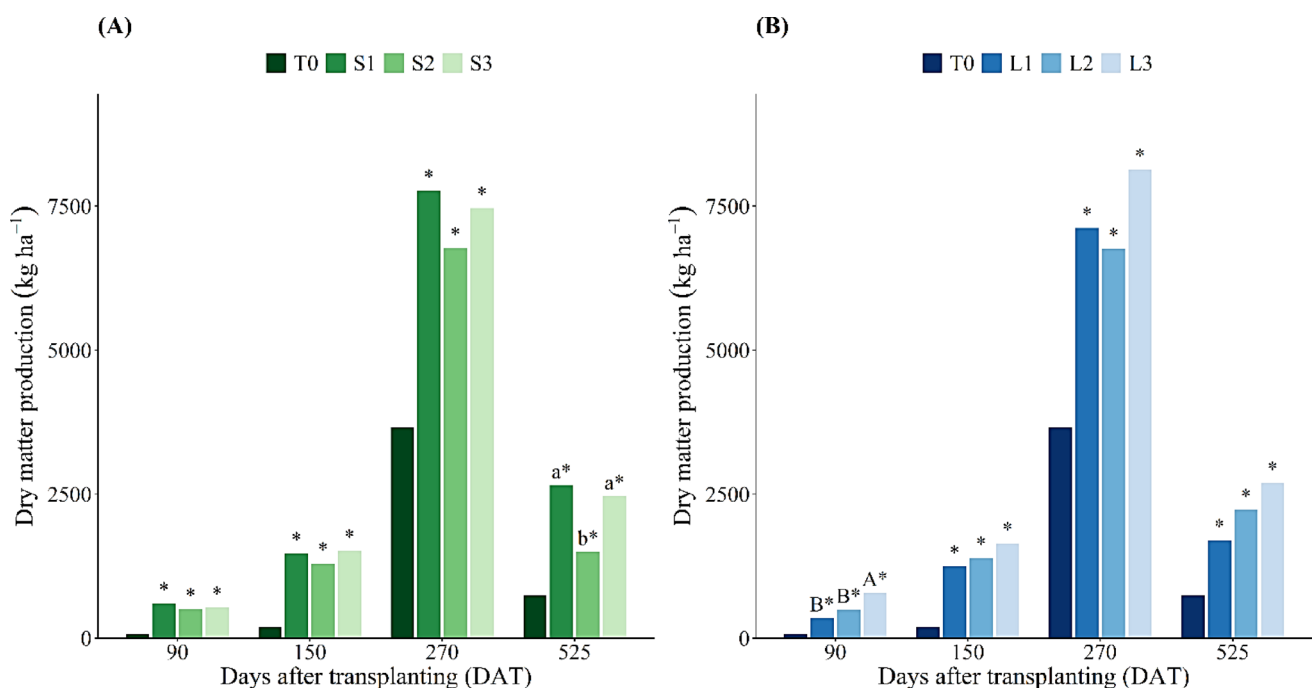
because carbon allocation and tissue ion concentrations were not measured, this explanation remains inferential.

For irrigation level (Fig. 7B), 100% ETc increased dry matter at 90 DAT by 40% and 125% relative to 33% and 66% ETc, respectively. In later evaluations, differences among irrigation levels were reduced, suggesting partial growth compensation after the early establishment phase. Such responses are compatible with dynamic allocation between growth and stress acclimation under changing water availability (Aanniz et al. 2025).

Table 6 shows significant salinity effects on seed oil content, seed yield, and oil yield, and significant irrigation-level effects on fruit, seed, and oil yields. Several yield variables also differed between irrigated treatments and the non-irrigated control.

Plants in S2 had the highest seed oil content (35.1%), but this percentage did not translate into the greatest oil yield; oil yield under S2 was 126.3 kg ha<sup>-1</sup> (Fig. 8). This distinction is expected because seed oil content represents the proportion of oil in processed seed, whereas oil yield integrates both seed production and oil concentration. Across treatments, seed oil content averaged approximately 33%.

The mean seed oil content obtained in this study (approximately 33%) was slightly below the 35–45% range reported for moringa seeds from other environments (Ayerza 2019), illustrating that oil concentration varies with genotype and growing conditions. The highest fruit, seed, and oil yields



**Fig. 7** Moringa shoot dry matter at different evaluation periods as affected by salinity of simulated oilfield-produced water (A) and irrigation level (B). \*Significant difference from the non-irrigated control (T0) by Dunnett's test at 5% probability. Identical lowercase letters

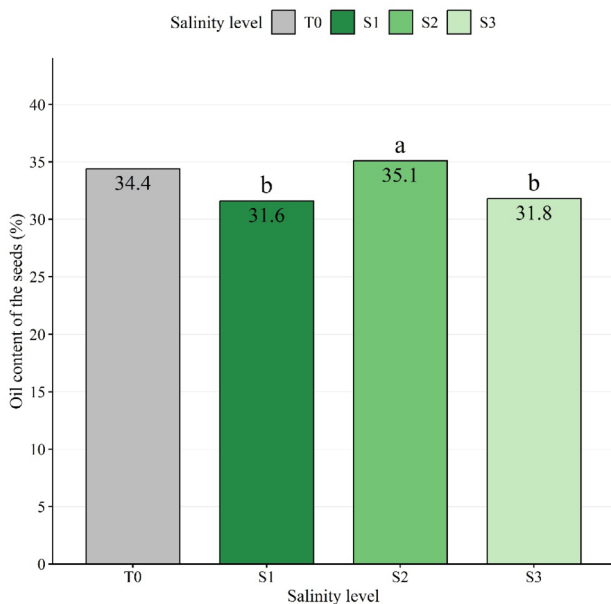
indicate no significant difference among salinity levels, and identical uppercase letters indicate no significant difference among irrigation levels by Tukey's test ( $p \leq 0.05$ )

**Table 6** Analysis of variance for seed oil content (SOC), fruit yield (FY), seed yield (SY), and oil yield (OY) of moringa under irrigation levels (Irr) and salinity levels (Sal) of simulated oilfield-produced water

| SV                             | DF   | SOC                 | FY                  | SY                  | OY                  |
|--------------------------------|------|---------------------|---------------------|---------------------|---------------------|
|                                |      | %                   | kg ha <sup>-1</sup> |                     |                     |
|                                |      | F-test significance |                     |                     |                     |
| Fat × T0                       | 1    | 0.392 <sup>ns</sup> | 0.001**             | 0.001**             | 0.002**             |
| Sal                            | 2    | 0.040*              | 0.058 <sup>ns</sup> | 0.005**             | 0.047*              |
| Bla                            | 2    | 0.661 <sup>ns</sup> | 0.001**             | 0.020*              | 0.030*              |
| Sal × Bla                      | 4    | 0.569 <sup>ns</sup> | 0.509 <sup>ns</sup> | 0.551 <sup>ns</sup> | 0.497 <sup>ns</sup> |
| dms(Sal or Bla)                | 3.4  | 376.5               | 131.6               | 38.8                |                     |
| dms(Dunnett) T0 vs. Sal or Bla | 4.6  | 500.9               | 175.1               | 51.6                |                     |
| dms (Dunnett) T0 vs. Trat      | 6.89 | 746.2               | 260.78              | 76.83               |                     |
| CV (%)                         | 10.4 | 27.2                | 27.3                | 26.8                |                     |
| Means                          |      | 33.0                | 1365.6              | 475.2               | 142.7               |

SV Source of variation, DF Degrees of freedom, DAT Days after transplanting, Fat Means of irrigated treatments, LSD Least significant difference, CV Coefficient of variation, T0 Non-irrigated control treatment, ns Not significant

\*Significant at the 5% probability level; \*\*Significant at the 1% probability level

**Fig. 8** Seed oil content of *Moringa oleifera* under salinity levels of simulated oilfield-produced water and irrigation levels. \*Significant difference from the non-irrigated control (T0) by Dunnett's test at 5% probability. Identical lowercase letters indicate no significant difference among salinity levels, and identical uppercase letters indicate no significant difference among irrigation levels by Tukey's test ( $p \leq 0.05$ )

occurred under both 66% ETc (L2) and 100% ETc (L3). Accordingly, L2 should be interpreted as a water-saving agronomic compromise rather than as an absolute physiological optimum: full irrigation promoted greater vegetative growth, whereas 66% ETc maintained high reproductive yield with a smaller irrigation input.

The 4.5 dS m<sup>-1</sup> treatment (S3) produced higher fruit, seed, and oil yields than the lower-salinity treatments (Fig. 9A), despite its greater electrical conductivity. One plausible explanation is that moringa's moderate salt tolerance interacted with the Ca<sup>2+</sup>, Mg<sup>2+</sup>, and K<sup>+</sup> supplied in the synthetic solution, which may have partially offset Na<sup>+</sup>-related ionic imbalance. Magnesium, in particular, is central to photosynthesis, assimilate transport, and seed filling in oilseed crops (Geng et al. 2021; Li et al. 2023; Wang et al. 2020), and Mg nutrition can contribute to drought tolerance (Santos et al. 2023). Nevertheless, plant nutrient uptake and tissue ion composition were not measured; therefore, the present experiment cannot distinguish a nutritional contribution from other salinity-acclimation mechanisms.

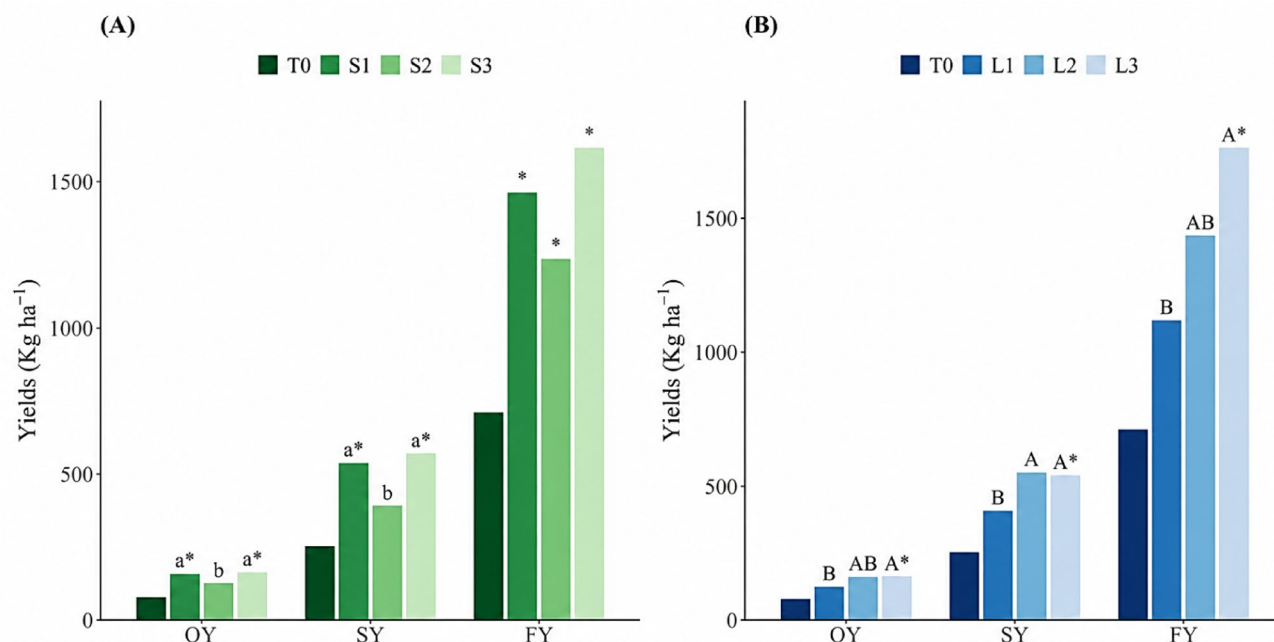
Table 7 summarizes crude-oil quality. Turbidity, acidity index, and kinematic viscosity were not significantly affected by salinity or irrigation level ( $p > 0.05$ ), whereas density showed a significant contrast between irrigated treatments and the non-irrigated control. Density is relevant to fuel handling and combustion and varies with feedstock composition and temperature (Ramírez-Verduzco 2022). However, the present measurements characterize crude moringa oil rather than finished biodiesel; therefore, engine-performance implications should not be inferred directly from these data.

Mean turbidity, acidity index, and kinematic viscosity were 5.0 NTU, 6.1 mg KOH g<sup>-1</sup>, and 41.3 mm<sup>2</sup> s<sup>-1</sup>, respectively. Because these variables did not differ significantly among the salinity or irrigation treatments, the tested irrigation factors did not produce a detectable change in these crude-oil quality attributes. This finding concerns only the agronomic treatments evaluated and should not be interpreted as evidence that irrigation with actual produced water would leave oil quality unaffected.

Oil turbidity averaged 5.0 NTU, but the coefficient of variation was high (242.6%), indicating substantial variability among samples. Turbidity reflects suspended or colloidal material and can be reduced during oil purification; therefore, it is more appropriately interpreted here as a processing-quality indicator than as evidence of irrigation-water safety (Pereira et al. 2016; Guerrini et al. 2020).

For biodiesel production, the crude oil would require subsequent processing and evaluation against the applicable fuel-quality specifications. The present study did not produce or test finished biodiesel; consequently, no direct conclusion about regulatory compliance or engine performance can be drawn from the crude-oil measurements alone (ANP 2024).

The mean acidity index was 6.1 mg KOH g<sup>-1</sup>. This relatively high value indicates that pretreatment would be required before conventional base-catalyzed transesterification because free fatty acids can promote saponification



**Fig. 9** Oil yield (OY), seed yield (SY), and fruit yield (FY) of *Moringa oleifera* as affected by salinity of simulated oilfield-produced water (A) and irrigation level (B). \*Significant difference from the non-irrigated control (T0) by Dunnett's test at 5% probability. Identical low-

er case letters indicate no significant difference among salinity levels, and identical uppercase letters indicate no significant difference among irrigation levels by Tukey's test ( $p \leq 0.05$ )

**Table 7** Analysis of variance for qualitative characteristics of crude moringa oil under irrigation levels (Irr) and salinity levels (Sal) of simulated oilfield-produced water

| SV                              | DF    | Turbidity           | Acidity index             | Kinematic viscosity             | Density             |
|---------------------------------|-------|---------------------|---------------------------|---------------------------------|---------------------|
|                                 |       | (NTU)               | (mg KOH g <sup>-1</sup> ) | mm <sup>2</sup> s <sup>-1</sup> | g cm <sup>-3</sup>  |
| F-test significance             |       |                     |                           |                                 |                     |
| Fat × T0                        | 1     | 0.550 <sup>ns</sup> | 0.367 <sup>ns</sup>       | 0.081 <sup>ns</sup>             | 0.039*              |
| Sal                             | 2     | 0.921 <sup>ns</sup> | 0.078 <sup>ns</sup>       | 0.427 <sup>ns</sup>             | 0.592 <sup>ns</sup> |
| Bla                             | 2     | 0.307 <sup>ns</sup> | 0.186 <sup>ns</sup>       | 0.636 <sup>ns</sup>             | 0.422 <sup>ns</sup> |
| Sal × Bla                       | 4     | 0.167 <sup>ns</sup> | 0.261 <sup>ns</sup>       | 0.668 <sup>ns</sup>             | 0.609 <sup>ns</sup> |
| dms (Sal or Bla)                | 12.37 | 5.38                | 2.97                      | 0.002                           |                     |
| dms (Dunnett) T0 vs. Sal or Bla | 16.46 | 7.15                | 3.95                      | 0.003                           |                     |
| dms (Dunnett) T0 vs. Trat       | 24.52 | 10.65               | 5.88                      | 0.005                           |                     |
| CV (%)                          | 242.6 | 86.9                | 17.1                      | 15.8                            |                     |
| Means                           | 5.0   | 6.1                 | 41.3                      | 0.896                           |                     |

and reduce conversion efficiency. Seed moisture and storage conditions can also influence hydrolytic deterioration and oil acidity (Wiltshire et al. 2022).

Accordingly, strategies such as seed pre-drying, acid esterification, or oil refining may be needed before biodiesel conversion of high-acidity moringa oil (Leone et al. 2016; Pereira et al. 2016; Omonhinmin et al. 2020). These processing requirements are independent of the central irrigation

result and should be considered when evaluating moringa oil as a bioenergy feedstock.

From an agronomic perspective, the results show that moringa maintained growth and reproductive performance when irrigated with synthetic saline waters up to 4.5 dS m<sup>-1</sup> under the conditions of this experiment. This finding is relevant to the development of water-saving strategies in semi-arid regions, but it represents a salinity-and-major-ion simulation rather than a field demonstration of agricultural reuse of actual oilfield-produced water.

The environmental implications therefore require caution. Repeated irrigation with saline water can increase soil electrical conductivity and sodium accumulation and may alter soil structure, infiltration, nutrient availability, and leaching dynamics. In this experiment, post-treatment soil salinity, sodicity, trace metals, organic contaminants, and drainage-water quality were not monitored, and the 708 mm of effective rainfall may have reduced salt accumulation during the rainy season. Moreover, actual produced water may contain hydrocarbons, dissolved organic compounds, trace metals, and treatment chemicals that were absent from the synthetic solutions. Before any practical reuse is recommended, experiments with appropriately treated real produced water should include comprehensive water characterization and long-term monitoring of soil EC and SAR, major ions, trace metals, organic contaminants, crop tissues, leachate, and potential groundwater impacts.

## Conclusions

Under the conditions of this field experiment, *Moringa oleifera* tolerated synthetic saline waters with electrical conductivities of 0.5–4.5 dS m<sup>-1</sup> formulated to simulate the major-ion composition of treated oilfield-produced water. Salinity within this range did not significantly reduce plant height or stem diameter among irrigated treatments, and the 4.5 dS m<sup>-1</sup> treatment was associated with high seed and oil yields. Both 66% and 100% ETc produced high reproductive yields; therefore, 66% ETc represents a water-saving agronomic compromise, whereas 100% ETc favored greater vegetative growth.

These results apply only to the synthetic-water formulation and experimental period evaluated. They do not demonstrate the safety or long-term feasibility of irrigating with actual oilfield-produced water, because residual hydrocarbons, organic compounds, trace metals, treatment chemicals, and long-term effects on soil salinity and sodicity were not assessed. Validation with treated real produced water and long-term soil, water, and crop monitoring is required before practical agricultural reuse can be recommended.

**Author contributions** Jéssica Crhistie de Castro Granjeiro Oliveira: Conceptualization, Methodology, Investigation, Formal analysis and data curation, Supervision, Writing—original draft preparation and Writing—review and editing. Francielle Gurgel de Castro Alves: Conceptualization, Methodology, Investigation, Formal analysis and data curation, Supervision, Writing—original draft preparation and Writing—review and editing. José Francismar de Medeiros: Conceptualization, Methodology, Investigation, Formal analysis and data curation, Supervision, Writing—original draft preparation and Writing—review and editing. Rodrigo Rafael da Silva: Conceptualization, Methodology, Investigation, Formal analysis and data curation, Supervision, Writing—original draft preparation and Writing—review and editing. Francimar Maik da Silva Morais: Conceptualization, Methodology, Investigation, Formal analysis and data curation, Supervision, Writing—original draft preparation and Writing—review and editing. Mateus de Freitas Almeida dos Santos: Conceptualization, Methodology, Investigation, Formal analysis and data curation, Supervision, Writing—original draft preparation and Writing—review and editing. Frederico Ribeiro do Carmo: Methodology, Investigation, Supervision, Writing—original draft preparation and Writing—review and editing. Luiz Fernando de Sousa Antunes: Writing—review and editing.

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**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** The authors declare no competing interests.

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